

Contrast Media Reaction Management

Policy Number	POL-CLN-045	Owner / Department	Radiology / Nursing
Effective Date	2023-10-02	Version	2.4
Last Revised	2026-02-05	Next Review	2028-02-05
Approved By	Patient Safety & Quality Committee	Applies To	Radiology technologists, nurses, and physicians involved in contrast-enhanced imaging

1. Purpose

To standardize the recognition and treatment of acute adverse reactions to iodinated and gadolinium-based contrast media, following the American College of Radiology (ACR) Manual on Contrast Media.

2. Scope

Applies to all areas where intravascular contrast is administered. A provider able to manage an acute reaction and an emergency cart with epinephrine must be immediately available whenever contrast is injected.

3. Screening and Prevention

- Screen for prior contrast reaction, asthma, and renal impairment (eGFR) before injection.
- A prior moderate or severe allergic-like reaction is the strongest predictor of a future reaction; consider a premedication regimen (corticosteroid plus antihistamine) and/or an alternative study.
- Confirm two patient identifiers, agent, dose, and IV patency before injection.

4. Reaction Severity and Treatment

IMPORTANT — For any severe allergic-like reaction — laryngeal edema with stridor, bronchospasm with hypoxia, or anaphylactic shock — give intramuscular epinephrine 0.3 mg (0.3 mL of 1 mg/mL) in the mid-outer thigh immediately, call a Code Blue (dial 55), give high-flow oxygen, and run IV fluids. Do not delay epinephrine to try antihistamines first.

Severity	Signs	First-line treatment
Mild	Limited urticaria/pruritus, nasal congestion, sneezing, limited scratchy throat; self-limited.	Observe and reassure; oral/IV antihistamine if bothersome.
Moderate	Diffuse urticaria, facial edema without dyspnea, wheezing/bronchospasm with mild or no hypoxia, stable vitals.	Oxygen, IV fluids, antihistamine, inhaled beta-agonist; monitor closely for progression.
Severe	Laryngeal edema with stridor, bronchospasm with hypoxia, diffuse erythema with hypotension, anaphylactic shock.	IM epinephrine 0.3 mg, high-flow oxygen, rapid IV fluids; call Code Blue.

Treat contrast extravasation by stopping the injection, elevating the limb, and applying cold compresses; escalate to surgical consult for compartment syndrome, skin ulceration, or a large-volume extravasation.

5. Post-Event Observation

- Observe patients who received epinephrine for a protracted period and arrange appropriate monitoring for biphasic reactions.
- Document the agent, dose, reaction, treatment, times, and outcome.
- Flag the reaction prominently in the allergy record.

6. Severity & Reporting

Report contrast reactions and management per POL-RM-013.

Severity	Example	Notify / timeframe
SEV-1	Severe reaction / anaphylaxis with cardiac arrest, death, or permanent harm; contrast given despite a documented severe prior reaction (sentinel event).	Attending, Administrator-on-Call, and Risk Management within 1 hour.
SEV-2	Moderate/severe reaction requiring rescue treatment with major temporary harm, or significant extravasation.	Radiologist and Risk Management within 4 hours.
SEV-3	Mild self-limited reaction; minor extravasation with no harm.	Radiology supervisor within 24 hours via event report.
Near Miss	High-risk allergy history caught before injection; wrong agent intercepted.	Event report within 24 hours.

7. References

- American College of Radiology, ACR Manual on Contrast Media (acute reaction classification and treatment).
- Related: POL-CLN-040, POL-CLN-046, POL-RM-013.